Low-Level Design (LLD) — CredVault

System: CredVault Credit Card Management Platform
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1. Module Breakdown

Every service follows Clean Architecture with four layers:

```
API Layer (Controllers, Middleware, DI)
   ↓
Application Layer (Commands, Queries, DTOs, Validators, Handlers)
   ↓
Infrastructure Layer (EF Core DbContext, Repositories, Messaging, External Clients)
   ↓
Domain Layer (Entities, Enums, Domain Events)
```

Dependency Rule: Inner layers never depend on outer layers.

1.1 Identity Service (:5001)

Domain Entity: `IdentityUser`

Commands:

Command	Handler Action
`RegisterUserCommand`	Create user (PendingVerification), generate OTP, publish `IUserRegistered` + `IUserOtpGenerated`
`LoginUserCommand`	Validate credentials, check status, return JWT
`GoogleLoginCommand`	Validate Google IdToken, create/login user, return JWT
`VerifyEmailOtpCommand`	Validate OTP, mark user Active, return JWT
`ForgotPasswordCommand`	Generate reset OTP, publish `IUserOtpGenerated`
`ResetPasswordCommand`	Validate OTP, update password hash
`UpdateUserProfileCommand`	Update FullName
`ChangePasswordCommand`	Validate old password, set new hash
`AdminUpdateUserStatusCommand`	Change user status (Active/Suspended)
`AdminUpdateUserRoleCommand`	Change user role (User/Admin)

Queries: `GetUserProfileQuery`, `AdminListUsersQuery`, `AdminGetUserQuery`, `AdminGetUserStatsQuery`

Events Published: `IUserRegistered`, `IUserOtpGenerated`

Infrastructure: `IdentityDbContext` (1 table), auto-migration, `UserRepository`.

1.2 Card Service (:5002)

Domain Entities: `CreditCard`, `CardTransaction`, `CardIssuer`

Commands: `AddCardCommand` (encrypt, mask, publish `ICardAdded`), `UpdateCardCommand`, `DeleteCardCommand` (soft-delete), `SetDefaultCardCommand`, `AddCardTransactionCommand`, `AdminUpdateCardCommand`, Issuer CRUD commands.

Queries: `GetCardsByUserQuery` (excludes soft-deleted), `GetCardByIdQuery`, `GetCardTransactionsQuery`, `GetAllUserTransactionsQuery`, `ListIssuersQuery`.

Events Published: `ICardAdded`

Infrastructure: `CardDbContext` (3 tables), AES encryption service, soft-delete global query filter.

1.3 Billing Service (:5003)

Domain Entities: `Bill`, `Statement`, `StatementTransaction`, `RewardAccount`, `RewardTier`, `RewardTransaction`

Commands: `GenerateBillCommand`, `CheckOverdueBillsCommand`, `PayBillCommand`, `CreateStatementCommand`, `EarnRewardsCommand`, `RedeemRewardsCommand` (saga step), `ReverseRewardsCommand` (saga compensation), RewardTier CRUD.

Queries: `GetBillsByUserQuery`, `GetBillByIdQuery`, `HasPendingBillQuery`, `GetStatementsByUserQuery`, `GetStatementByIdQuery`, `GetRewardAccountQuery`, `ListRewardTiersQuery`.

Events Consumed: `IBillUpdateRequested`, `IRewardRedemptionRequested`, `IRevertBillUpdate`, `IReverseRewards` (from Payment Service saga).

Infrastructure: `BillingDbContext` (6 tables), statement aggregation, tier-based reward calculation.

1.4 Payment Service (:5004)

Domain Entities: `Payment`, `Transaction`, `UserWallet`, `WalletTransaction`, `PaymentOrchestrationSagaState`, `RazorpayWalletTopUp`

Commands: `InitiatePaymentCommand` (validate, create, OTP, publish), `VerifyPaymentOtpCommand` (triggers saga), `TopUpWalletCommand` (Razorpay order), `ProcessRazorpayWebhookCommand`, `DebitWalletCommand` (saga step), `RefundWalletCommand` (compensation).

Queries: `GetPaymentByIdQuery`, `GetWalletQuery`, `GetWalletBalanceQuery`, `GetWalletTransactionsQuery`.

Saga: `PaymentOrchestrationSaga` (MassTransit State Machine) — see Section 5.4.

Events Published: `IPaymentOtpGenerated`, `IStartPaymentOrchestration`, saga request/response events, `IPaymentCompleted`, `IPaymentCompensated`.

Background Jobs: `PaymentExpirationJob` — expires unpaid payments.

Infrastructure: `PaymentDbContext` (6 tables), Razorpay client, saga state repository.

1.5 Notification Service (:5005)

Domain Entities: `AuditLog`, `NotificationLog`

Consumers (MassTransit):

Event	Action
`IUserRegistered`	Send welcome email
`IUserOtpGenerated`	Send OTP email (verification or password reset)
`ICardAdded`	Send card confirmation email
`IPaymentOtpGenerated`	Send payment OTP email
`IPaymentCompleted`	Send payment success email
`IPaymentCompensated`	Send payment failure email

Commands: `SendEmailCommand`, `LogAuditCommand`, `LogNotificationCommand`

Queries: `GetNotificationLogsQuery`, `GetAuditLogsQuery`

Infrastructure: `NotificationDbContext` (2 tables), `GmailSmtpEmailSender`, retry policy (1s → 5s → 15s).

2. Class & Schema Design

2.1 Domain Entities Summary

Entity	Service	Key Fields
`IdentityUser`	Identity	Id, Email (UQ), FullName, PasswordHash (nullable), IsEmailVerified, EmailVerificationOtp, PasswordResetOtp, Status (enum), Role (enum), CreatedAtUtc, UpdatedAtUtc
`CreditCard`	Card	Id, UserId (indexed), IssuerId (FK), CardholderName, Last4, MaskedNumber, EncryptedCardNumber, ExpMonth, ExpYear, CreditLimit, OutstandingBalance, IsDefault, IsVerified, IsDeleted (soft), timestamps
`CardTransaction`	Card	Id, CardId (FK), UserId, Type (enum), Amount, Description, DateUtc
`CardIssuer`	Card	Id, Name, Network (enum), timestamps
`Bill`	Billing	Id, UserId, CardId, CardNetwork, IssuerId, Amount, MinDue, Currency, BillingDateUtc, DueDateUtc, AmountPaid, PaidAtUtc, Status (enum), timestamps
`Statement`	Billing	Id, UserId, CardId, BillId (FK), Period, balances (opening/purchases/payments/refunds/penalties/interest/closing), status, card metadata, timestamps
`StatementTransaction`	Billing	Id, StatementId (FK), SourceTransactionId, Type, Amount, Description, DateUtc
`RewardAccount`	Billing	Id, UserId (UQ), RewardTierId (FK), PointsBalance, timestamps
`RewardTier`	Billing	Id, CardNetwork, IssuerId (nullable), MinSpend, RewardRate, EffectiveFromUtc, EffectiveToUtc, timestamps
`RewardTransaction`	Billing	Id, RewardAccountId (FK), BillId (FK, nullable), Points, Type (Earned/Redeemed/Reversed), CreatedAtUtc
`Payment`	Payment	Id, UserId, CardId, BillId, Amount, PaymentType (enum), Status (enum), FailureReason, OtpCode, OtpExpiresAtUtc, timestamps
`Transaction`	Payment	Id, PaymentId (FK), UserId, Amount, Type (Debit/Credit), Description, timestamps
`UserWallet`	Payment	Id, UserId (UQ), Balance, TotalTopUps, TotalSpent, timestamps
`WalletTransaction`	Payment	Id, WalletId (FK), Type (TopUp/Debit/Refund), Amount, BalanceAfter, Description, RelatedPaymentId, CreatedAtUtc
`PaymentOrchestrationSagaState`	Payment	CorrelationId (PK), CurrentState, UserId, CardId, BillId, Email, FullName, Amount, RewardsAmount, PaymentType,

Entity	Service	Key Fields
		CompensationReason, error fields
`RazorpayWalletTopUp`	Payment	Id, UserId, Amount, RazorpayOrderId (UQ), RazorpayPaymentId (UQ), RazorpaySignature, Status, FailureReason, timestamps
`AuditLog`	Notification	Id, EntityName, EntityId, Action, UserId, Changes (JSON), TraceId, CreatedAtUtc
`NotificationLog`	Notification	Id, UserId, Recipient, Subject, Body, Type, IsSuccess, ErrorMessage, TraceId, CreatedAtUtc

2.2 Application Layer Pattern

CQRS via MediatR:

- Commands mutate state (create, update, delete) — return result objects
- Queries read data (fetch, list, search) — return DTOs
- Each handler is isolated — no shared logic between reads and writes
- FluentValidation runs as a pipeline behavior before handlers execute

Shared Contracts Library (`Shared.Contracts`):

- `BaseApiController` — standardized response methods (Ok, Created, BadRequest, etc.)
- `ApiResponse<T>` — consistent API envelope (success, data, message, errors)
- `ExceptionHandlingMiddleware` — global error handling, returns 500 envelope
- `ServiceCollectionExtensions` — JWT, Swagger, MassTransit DI setup
- All event interfaces and enum definitions shared across services

2.3 Event Contracts

Identity Events:

```
IUserRegistered:     userId, email, fullName, role, registeredAt
IUserOtpGenerated:   userId, email, otp, otpType (EmailVerification/PasswordReset), expiresAt
```

Card Events:

```
ICardAdded:          cardId, userId, cardLast4, cardNetwork, issuerName, addedAt
```

Payment Events:

```
IPaymentOtpGenerated:     paymentId, userId, email, otp, amount, expiresAt
IStartPaymentOrchestration: correlationId, paymentId, userId, cardId, billId, email, fullName, amount, re
```

Saga Events:

```
IBillUpdateRequested:     correlationId, billId, amount, paymentId
IBillUpdated:             correlationId, billId, success
IRewardRedemptionRequested: correlationId, userId, rewardsAmount, billId
IRewardsRedeemed:         correlationId, pointsRedeemed, success
ICardDeductionRequested:  correlationId, cardId, amount
ICardDeducted:            correlationId, cardId, success
IPaymentCompleted:        correlationId, paymentId, completedAt
IPaymentCompensated:      correlationId, paymentId, compensationReason, compensatedAt
```

Event Routing:

Event	Publisher	Consumer
`IUserRegistered`	Identity	Notification
`IUserOtpGenerated`	Identity	Notification
`ICardAdded`	Card	Notification
`IPaymentOtpGenerated`	Payment	Notification
`IStartPaymentOrchestration`	Payment	Payment (Saga)
`IBillUpdateRequested`	Payment (Saga)	Billing
`IBillUpdated`	Billing	Payment (Saga)
`IRewardRedemptionRequested`	Payment (Saga)	Billing
`IRewardsRedeemed`	Billing	Payment (Saga)
`ICardDeductionRequested`	Payment (Saga)	Card
`ICardDeducted`	Card	Payment (Saga)
`IPaymentCompleted`	Payment (Saga)	Notification
`IPaymentCompensated`	Payment (Saga)	Notification

3. Database Schema (Detailed)

3.1 credvault_identity

Table: identity_users

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
Email	NVARCHAR(256)	UQ, NOT NULL	User email

Column	Type	Constraints	Description
FullName	NVARCHAR(200)	NOT NULL	Display name
PasswordHash	NVARCHAR(512)	NULLABLE	BCrypt hash (null for SSO)
IsEmailVerified	BIT	NOT NULL, DEFAULT 0	Email verification flag
EmailVerificationOtp	NVARCHAR(6)	NULLABLE	Current OTP code
PasswordResetOtp	NVARCHAR(6)	NULLABLE	Password reset OTP
Status	INT	NOT NULL, DEFAULT 0	UserStatus enum
Role	INT	NOT NULL, DEFAULT 0	UserRole enum
CreatedAtUtc	DATETIME2	NOT NULL	Record creation time
UpdatedAtUtc	DATETIME2	NOT NULL	Last update time

Indexes: PK on `Id` , UQ on `Email`

3.2 credvault_cards

Table: CardIssuers

Column	Type	Constraints
Id	UNIQUEIDENTIFIER	PK, NOT NULL
Name	NVARCHAR(100)	NOT NULL
Network	INT	NOT NULL
CreatedAtUtc / UpdatedAtUtc	DATETIME2	NOT NULL

Table: CreditCards

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
UserId	UNIQUEIDENTIFIER	NOT NULL, INDEXED	Owner user ID
IssuerId	UNIQUEIDENTIFIER	FK → CardIssuers.Id	Card issuer
CardholderName	NVARCHAR(200)	NOT NULL	Name on card
Last4	NVARCHAR(4)	NOT NULL	Last 4 digits (plain)
MaskedNumber	NVARCHAR(19)	NOT NULL	**** **** **** XXXX
EncryptedCardNumber	NVARCHAR(512)	NOT NULL	AES-encrypted full number
ExpMonth	INT	NOT NULL	Expiry month (1-12)

Column	Type	Constraints	Description
ExpYear	INT	NOT NULL	Expiry year
CreditLimit	DECIMAL(18,2)	NOT NULL	Credit limit
OutstandingBalance	DECIMAL(18,2)	NOT NULL, DEFAULT 0	Current balance
BillingCycleStartDay	INT	NOT NULL	Day of month for billing
IsDefault	BIT	NOT NULL, DEFAULT 0	Default card flag
IsVerified	BIT	NOT NULL, DEFAULT 0	Verification status
IsDeleted	BIT	NOT NULL, DEFAULT 0	Soft delete flag
CreatedAtUtc / UpdatedAtUtc	DATETIME2	NOT NULL	Timestamps

Table: CardTransactions

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
CardId	UNIQUEIDENTIFIER	FK → CreditCards.Id	Related card
UserId	UNIQUEIDENTIFIER	NOT NULL	Owner user ID
Type	INT	NOT NULL	CardTransactionType enum
Amount	DECIMAL(18,2)	NOT NULL	Transaction amount
Description	NVARCHAR(500)	NULLABLE	Description
DateUtc	DATETIME2	NOT NULL	Transaction date

Relationships: CardIssuers 1→N CreditCards, CreditCards 1→N CardTransactions

3.3 credvault_billing

Table: Bills

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
UserId	UNIQUEIDENTIFIER	NOT NULL	Owner user ID
CardId	UNIQUEIDENTIFIER	NOT NULL	Related card
CardNetwork	INT	NOT NULL	CardNetwork enum
IssuerId	UNIQUEIDENTIFIER	NOT NULL	Card issuer
Amount	DECIMAL(18,2)	NOT NULL	Total bill amount

Column	Type	Constraints	Description
MinDue	DECIMAL(18,2)	NOT NULL	Minimum due amount
Currency	NVARCHAR(3)	NOT NULL, DEFAULT 'USD'	Currency code
BillingDateUtc	DATETIME2	NOT NULL	Bill generation date
DueDateUtc	DATETIME2	NOT NULL	Payment due date
AmountPaid	DECIMAL(18,2)	NOT NULL, DEFAULT 0	Amount paid so far
PaidAtUtc	DATETIME2	NULLABLE	Full payment date
Status	INT	NOT NULL, DEFAULT 0	BillStatus enum
CreatedAtUtc / UpdatedAtUtc	DATETIME2	NOT NULL	Timestamps

Table: Statements

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
UserId	UNIQUEIDENTIFIER	NOT NULL	Owner user ID
CardId	UNIQUEIDENTIFIER	NOT NULL	Related card
BillId	UNIQUEIDENTIFIER	FK → Bills.Id, NULLABLE	Related bill
StatementPeriod	NVARCHAR(50)	NOT NULL	Period label
PeriodStartUtc / PeriodEndUtc	DATETIME2	NOT NULL	Period range
OpeningBalance	DECIMAL(18,2)	NOT NULL	Starting balance
TotalPurchases / TotalPayments / TotalRefunds	DECIMAL(18,2)	NOT NULL	Aggregated amounts
PenaltyCharges / InterestCharges	DECIMAL(18,2)	NOT NULL, DEFAULT 0	Fees
ClosingBalance	DECIMAL(18,2)	NOT NULL	Final balance
MinimumDue	DECIMAL(18,2)	NOT NULL	Minimum payment
AmountPaid	DECIMAL(18,2)	NOT NULL, DEFAULT 0	Amount paid
Status	INT	NOT NULL, DEFAULT 0	StatementStatus enum
CardLast4 / CardNetwork / IssuerName	NVARCHAR	NOT NULL	Card metadata
CreditLimit / AvailableCredit	DECIMAL(18,2)	NOT NULL	Credit info

Column	Type	Constraints	Description
CreatedAtUtc / UpdatedAtUtc	DATETIME2	NOT NULL	Timestamps

Table: StatementTransactions

Column	Type	Constraints
Id	UNIQUEIDENTIFIER	PK, NOT NULL
StatementId	UNIQUEIDENTIFIER	FK → Statements.Id
SourceTransactionId	UNIQUEIDENTIFIER	NOT NULL
Type	INT	NOT NULL
Amount	DECIMAL(18,2)	NOT NULL
Description	NVARCHAR(500)	NULLABLE
DateUtc	DATETIME2	NOT NULL

Table: RewardAccounts

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
UserId	UNIQUEIDENTIFIER	UQ, NOT NULL	Owner (unique)
RewardTierId	UNIQUEIDENTIFIER	FK → RewardTiers.Id	Current tier
PointsBalance	DECIMAL(18,2)	NOT NULL, DEFAULT 0	Available points
CreatedAtUtc / UpdatedAtUtc	DATETIME2	NOT NULL	Timestamps

Table: RewardTiers

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
CardNetwork	INT	NOT NULL	CardNetwork enum
IssuerId	UNIQUEIDENTIFIER	NULLABLE	Specific issuer (null = all)
MinSpend	DECIMAL(18,2)	NOT NULL	Minimum spend
RewardRate	DECIMAL(18,4)	NOT NULL	Points per unit
EffectiveFromUtc	DATETIME2	NOT NULL	Effective start
EffectiveToUtc	DATETIME2	NULLABLE	Effective end
CreatedAtUtc / UpdatedAtUtc	DATETIME2	NOT NULL	Timestamps

Table: RewardTransactions

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
RewardAccountId	UNIQUEIDENTIFIER	FK → RewardAccounts.Id	Related account
BillId	UNIQUEIDENTIFIER	FK → Bills.Id, NULLABLE	Related bill
Points	DECIMAL(18,2)	NOT NULL	Points amount
Type	INT	NOT NULL	Earned / Redeemed / Reversed
CreatedAtUtc	DATETIME2	NOT NULL	Creation time

Relationships: Bills 1→N Statements, Statements 1→N StatementTransactions, RewardTiers 1→N RewardAccounts, RewardAccounts 1→N RewardTransactions, Bills 1→N RewardTransactions (nullable)

3.4 credvault_payments

Table: Payments

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
UserId	UNIQUEIDENTIFIER	NOT NULL	Owner user ID
CardId	UNIQUEIDENTIFIER	NOT NULL	Related card
BillId	UNIQUEIDENTIFIER	NOT NULL	Related bill
Amount	DECIMAL(18,2)	NOT NULL	Payment amount
PaymentType	INT	NOT NULL	PaymentType enum
Status	INT	NOT NULL, DEFAULT 0	PaymentStatus enum
FailureReason	NVARCHAR(500)	NULLABLE	Failure description
OtpCode	NVARCHAR(6)	NOT NULL	Payment OTP
OtpExpiresAtUtc	DATETIME2	NOT NULL	OTP expiry
CreatedAtUtc / UpdatedAtUtc	DATETIME2	NOT NULL	Timestamps

Table: Transactions

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
PaymentId	UNIQUEIDENTIFIER	FK → Payments.Id	Related payment
UserId	UNIQUEIDENTIFIER	NOT NULL	Owner user ID

Column	Type	Constraints	Description
Amount	DECIMAL(18,2)	NOT NULL	Transaction amount
Type	INT	NOT NULL	Debit / Credit
Description	NVARCHAR(500)	NULLABLE	Description
CreatedAtUtc / UpdatedAtUtc	DATETIME2	NOT NULL	Timestamps

Table: UserWallets

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
UserId	UNIQUEIDENTIFIER	UQ, NOT NULL	Owner (unique)
Balance	DECIMAL(18,2)	NOT NULL, DEFAULT 0	Current balance
TotalTopUps / TotalSpent	DECIMAL(18,2)	NOT NULL, DEFAULT 0	Lifetime totals
CreatedAtUtc / UpdatedAtUtc	DATETIME2	NOT NULL	Timestamps

Table: WalletTransactions

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
WalletId	UNIQUEIDENTIFIER	FK → UserWallets.Id	Related wallet
Type	INT	NOT NULL	TopUp / Debit / Refund
Amount	DECIMAL(18,2)	NOT NULL	Transaction amount
BalanceAfter	DECIMAL(18,2)	NOT NULL	Balance after transaction
Description	NVARCHAR(500)	NULLABLE	Description
RelatedPaymentId	UNIQUEIDENTIFIER	NULLABLE	Related payment
CreatedAtUtc	DATETIME2	NOT NULL	Creation time

Table: PaymentOrchestrationSagas

Column	Type	Constraints	Description
CorrelationId	UNIQUEIDENTIFIER	PK, NOT NULL	Saga correlation ID
CurrentState	NVARCHAR(50)	NOT NULL	Current saga state
UserId / CardId / BillId	UNIQUEIDENTIFIER	NOT NULL	Related entities
Email / FullName	NVARCHAR	NOT NULL	User details

Column	Type	Constraints	Description
Amount / RewardsAmount	DECIMAL(18,2)	NOT NULL	Payment amounts
PaymentType	INT	NOT NULL	PaymentType enum
CompensationReason	NVARCHAR(500)	NULLABLE	Compensation reason

Table: RazorpayWalletTopUps

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
UserId	UNIQUEIDENTIFIER	NOT NULL	Owner user ID
Amount	DECIMAL(18,2)	NOT NULL	Top-up amount
RazorpayOrderId	NVARCHAR(100)	UQ, NOT NULL	Razorpay order ID
RazorpayPaymentId	NVARCHAR(100)	UQ, NULLABLE	Razorpay payment ID
RazorpaySignature	NVARCHAR(512)	NULLABLE	Webhook signature
Status	INT	NOT NULL, DEFAULT 0	RazorpayTopUpStatus enum
FailureReason	NVARCHAR(500)	NULLABLE	Failure description
CreatedAtUtc / UpdatedAtUtc	DATETIME2	NOT NULL	Timestamps

Relationships: UserWallets 1→N WalletTransactions, Payments 1→N Transactions. Sagas and TopUps are independent tracking tables.

3.5 credvault_notifications

Table: AuditLogs

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
EntityName	NVARCHAR(100)	NOT NULL	Entity type
EntityId	UNIQUEIDENTIFIER	NOT NULL	Entity ID
Action	NVARCHAR(50)	NOT NULL	Action performed
UserId	UNIQUEIDENTIFIER	NOT NULL	Actor user ID
Changes	NVARCHAR(MAX)	NOT NULL	JSON change log
TraceId	NVARCHAR(100)	NOT NULL	Correlation trace ID
CreatedAtUtc	DATETIME2	NOT NULL	Creation time

Table: NotificationLogs

Column	Type	Constraints	Description
Id	UNIQUEIDENTIFIER	PK, NOT NULL	Primary key
UserId	UNIQUEIDENTIFIER	NULLABLE	Recipient user ID
Recipient	NVARCHAR(256)	NOT NULL	Email address
Subject	NVARCHAR(500)	NOT NULL	Email subject
Body	NVARCHAR(MAX)	NOT NULL	Email body
Type	NVARCHAR(10)	NOT NULL	NotificationType enum
IsSuccess	BIT	NOT NULL	Delivery success flag
ErrorMessage	NVARCHAR(1000)	NULLABLE	Error description
TraceId	NVARCHAR(100)	NOT NULL	Correlation trace ID
CreatedAtUtc	DATETIME2	NOT NULL	Creation time

3.6 Cross-Database Constraint Policy

No cross-service foreign key constraints. Referential integrity across services maintained through:

- Eventual consistency via RabbitMQ events
- Saga compensation for distributed transactions
- UserId/CardId/BillId passed as plain GUIDs in events and commands

4. API Design

4.1 Standard Response Format

All responses follow the `ApiResponse<T>` envelope via `BaseApiController`:

Success:

```
{
  "success": true,
  "data": { ... },
  "message": "Operation successful",
  "errors": []
}
```

Error:

```
{
  "success": false,
  "data": null,
  "message": "Validation failed",
  "errors": ["Email is required", "Password must be at least 8 characters"]
}
```

4.2 Identity Service (`/api/v1/identity`)

#	Method	Endpoint	Auth	Description
1	POST	`/auth/register`	Public	Register new user
2	POST	`/auth/login`	Public	Email/password login
3	POST	`/auth/google`	Public	Google OAuth login
4	POST	`/auth/verify-email-otp`	Public	Verify email with OTP
5	POST	`/auth/resend-verification`	Public	Resend verification OTP
6	POST	`/auth/forgot-password`	Public	Request password reset
7	POST	`/auth/reset-password`	Public	Reset password with OTP
8	GET	`/users/me`	User	Get current user profile
9	PUT	`/users/me`	User	Update current user profile
10	PUT	`/users/me/password`	User	Change password
11	GET	`/users`	Admin	List all users (paginated)
12	GET	`/users/{id}`	Admin	Get user by ID
13	PUT	`/users/{id}/status`	Admin	Update user status
14	PUT	`/users/{id}/role`	Admin	Update user role
15	GET	`/users/stats`	Admin	User statistics

4.3 Card Service (`/api/v1/cards`)

#	Method	Endpoint	Auth	Description
16	GET	`/cards`	User	List user's cards
17	POST	`/cards`	User	Add new card
18	GET	`/cards/{id}`	User	Get card details
19	PUT	`/cards/{id}`	User	Update card
20	PATCH	`/cards/{id}/default`	User	Set card as default

#	Method	Endpoint	Auth	Description
21	DELETE	`/cards/{id}`	User	Soft-delete card
22	GET	`/cards/{cardId}/transactions`	User	Get card transactions
23	POST	`/cards/{cardId}/transactions`	User	Record card transaction
24	GET	`/cards/transactions`	User	All transactions for user
25	GET	`/issuers`	User	List all issuers
26	GET	`/issuers/{id}`	User	Get issuer details
27	POST	`/issuers`	Admin	Create issuer
28	PUT	`/issuers/{id}`	Admin	Update issuer
29	DELETE	`/issuers/{id}`	Admin	Delete issuer

4.4 Billing Service (`/api/v1/billing`)

#	Method	Endpoint	Auth	Description
35	GET	`/bills`	User	List user's bills
36	GET	`/bills/{id}`	User	Get bill details
37	GET	`/bills/has-pending/{cardId}`	User	Check pending bill
38	POST	`/bills/admin/generate-bill`	Admin	Generate bill
39	POST	`/bills/admin/check-overdue`	Admin	Check overdue bills
40	GET	`/statements`	User	List user's statements
41	GET	`/statements/{id}`	User	Get statement details
42	GET	`/statements/{id}/transactions`	User	Get statement transactions
43	GET	`/rewards/account`	User	Get reward balance
44	GET	`/rewards/transactions`	User	Get reward history
45	POST	`/rewards/internal/redeem`	Internal	Redeem rewards (saga)
46	GET	`/rewards/tiers`	User	List reward tiers
47	POST	`/rewards/tiers`	Admin	Create reward tier
48	PUT	`/rewards/tiers/{id}`	Admin	Update reward tier
49	DELETE	`/rewards/tiers/{id}`	Admin	Delete reward tier

4.5 Payment Service (`/api/v1/payments`)

#	Method	Endpoint	Auth	Description
50	POST	`/payments/initiate`	User	Initiate payment (triggers OTP)
51	POST	`/payments/{id}/verify-otp`	User	Verify payment OTP
52	POST	`/payments/{id}/resend-otp`	User	Resend payment OTP
53	GET	`/payments/{id}`	User	Get payment details
54	GET	`/wallets/me`	User	Get wallet info
55	GET	`/wallets/balance`	User	Get wallet balance
56	GET	`/wallets/transactions`	User	Get wallet history
57	POST	`/wallets/topup`	User	Create Razorpay top-up order
58	POST	`/wallets/razorpay-webhook`	Public	Razorpay webhook callback

4.6 Notification Service (`/api/v1/notifications`)

#	Method	Endpoint	Auth	Description
59	GET	`/logs`	Admin	Get notification logs
60	GET	`/audit`	Admin	Get audit trail

4.7 Key Request DTOs

RegisterUserRequest — `POST /api/v1/identity/auth/register`

```
{
  "email": "string (required, valid email)",
  "fullName": "string (required, min 2 chars)",
  "password": "string (required, min 8 chars, 1 uppercase, 1 lowercase, 1 digit)"
}
```

AddCardRequest — `POST /api/v1/cards`

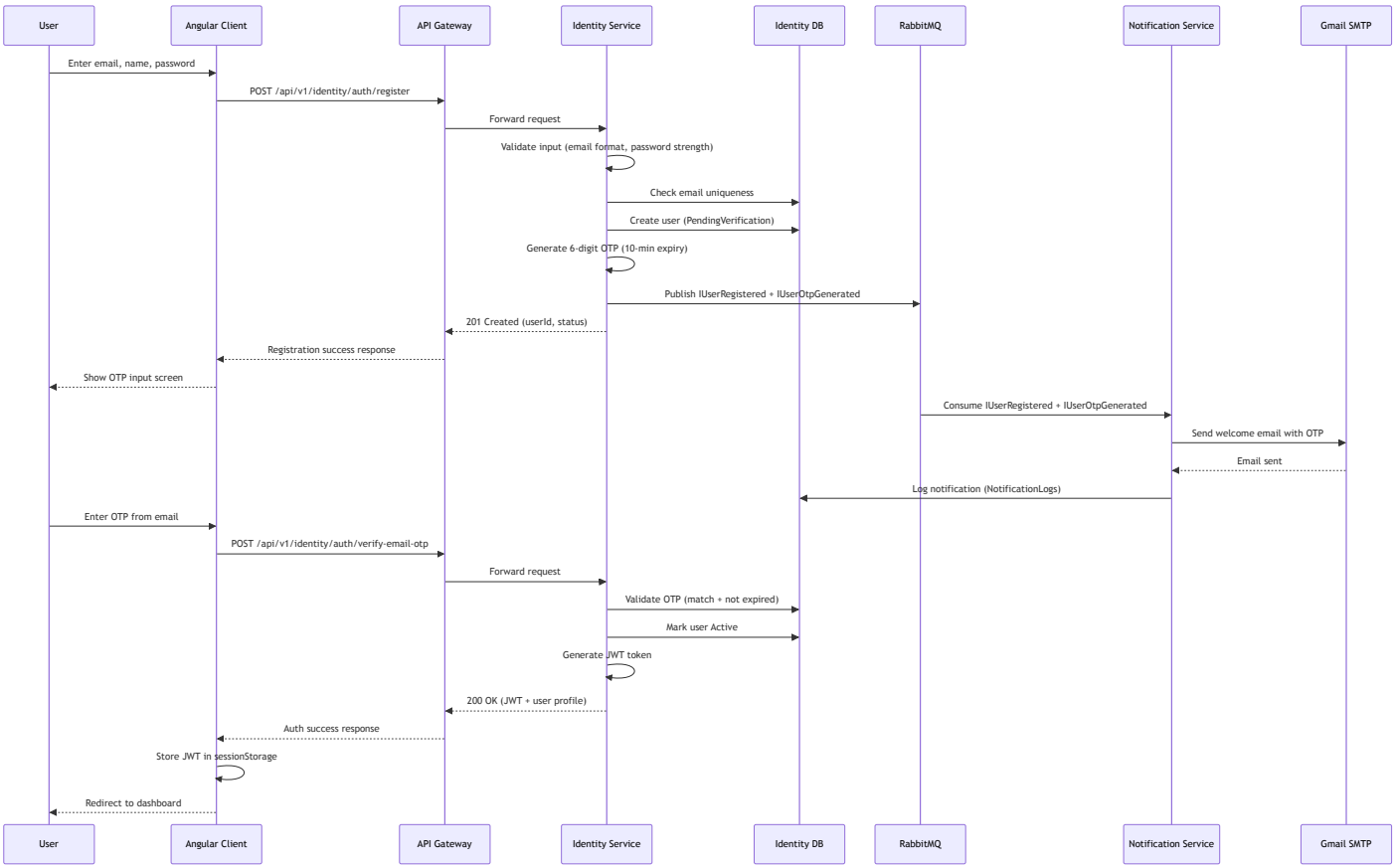
```
{
  "cardNumber": "string (required, 16 digits, Luhn valid)",
  "cardholderName": "string (required)",
  "expiryMonth": "int (required, 1-12)",
  "expiryYear": "int (required, >= current year)",
  "issuerId": "Guid (required)",
  "creditLimit": "decimal (required, > 0)",
  "billingCycleStartDay": "int (required, 1-28)"
}
```

InitiatePaymentRequest — `POST /api/v1/payments/initiate`

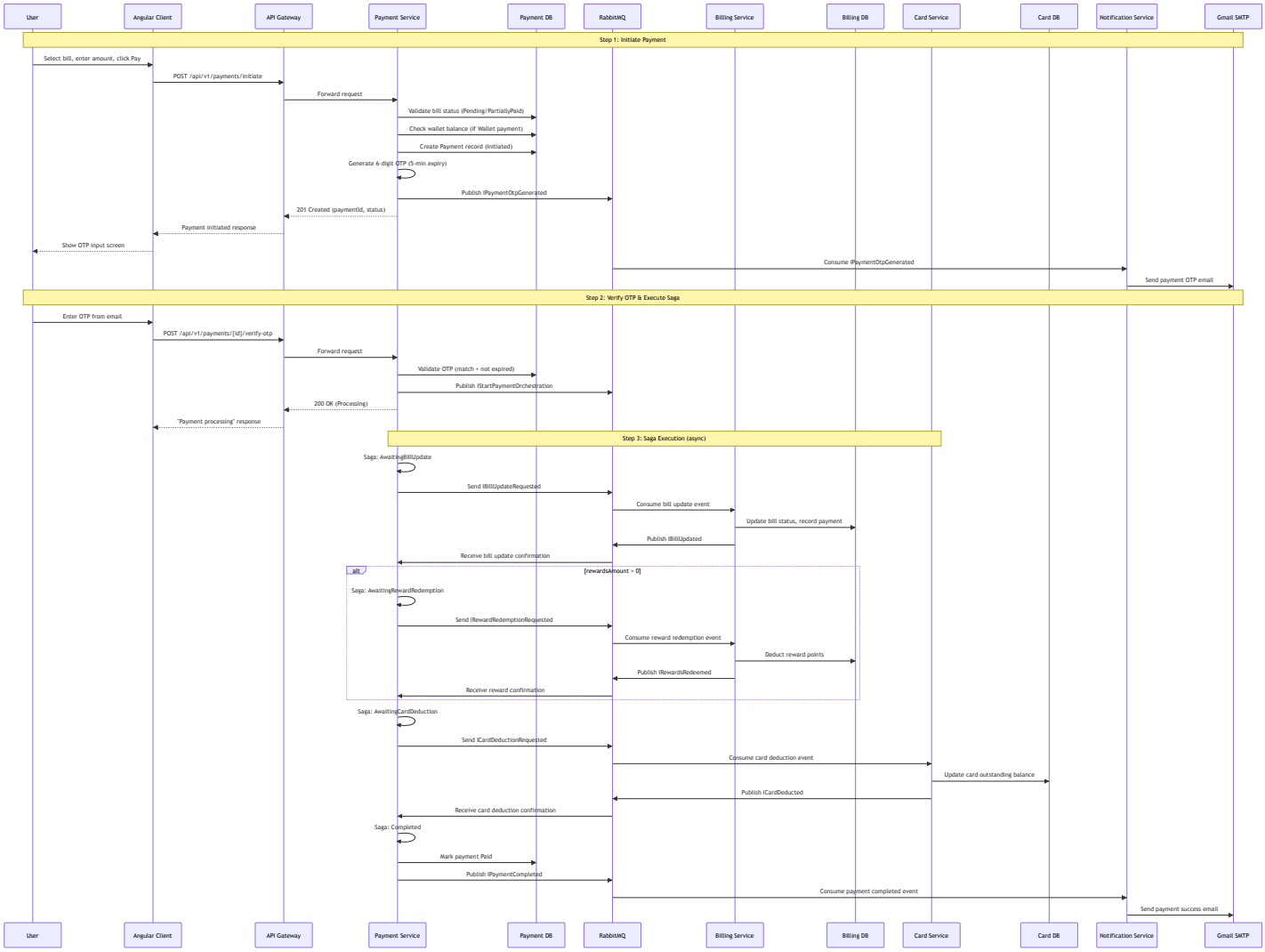
```
{
  "cardId": "Guid (required)",
  "billId": "Guid (required)",
  "amount": "decimal (required, > 0)",
  "paymentType": "int (required, 0=Wallet, 1=Card)",
  "rewardsAmount": "decimal (optional, 0 if not redeeming)"
}
```

5. Sequence Diagrams

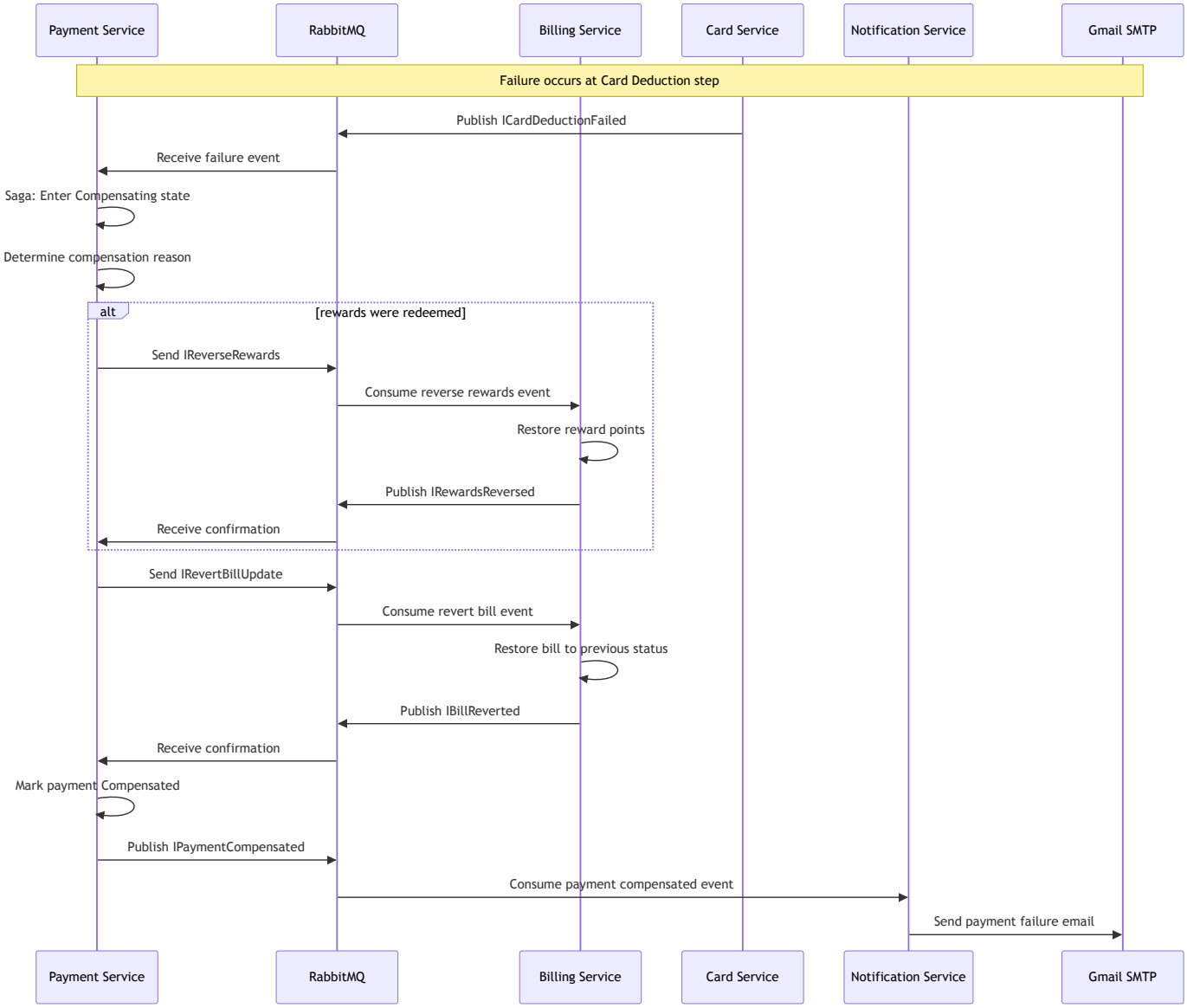
5.1 User Registration & Email Verification



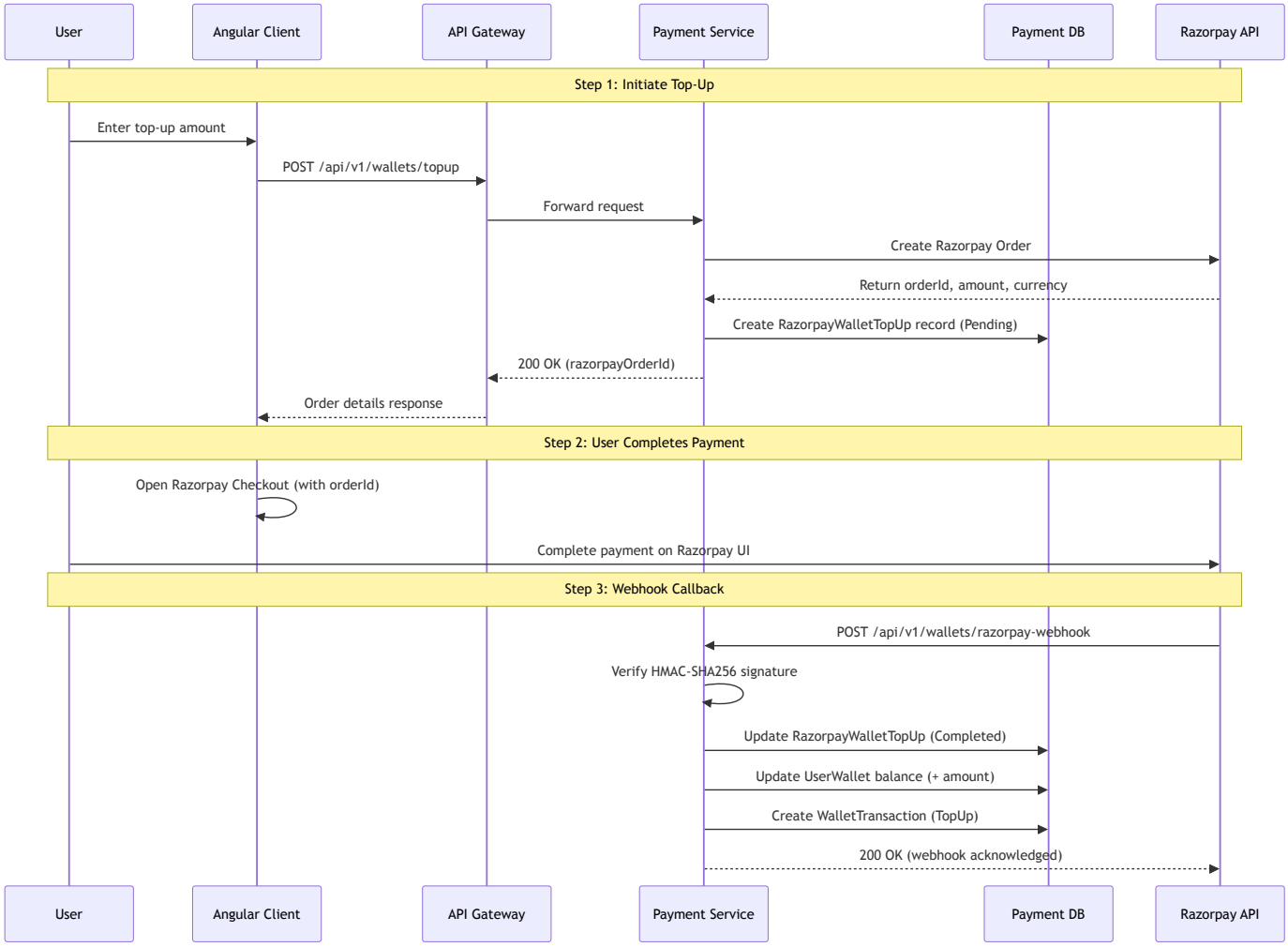
5.2 Bill Payment with Saga Orchestration



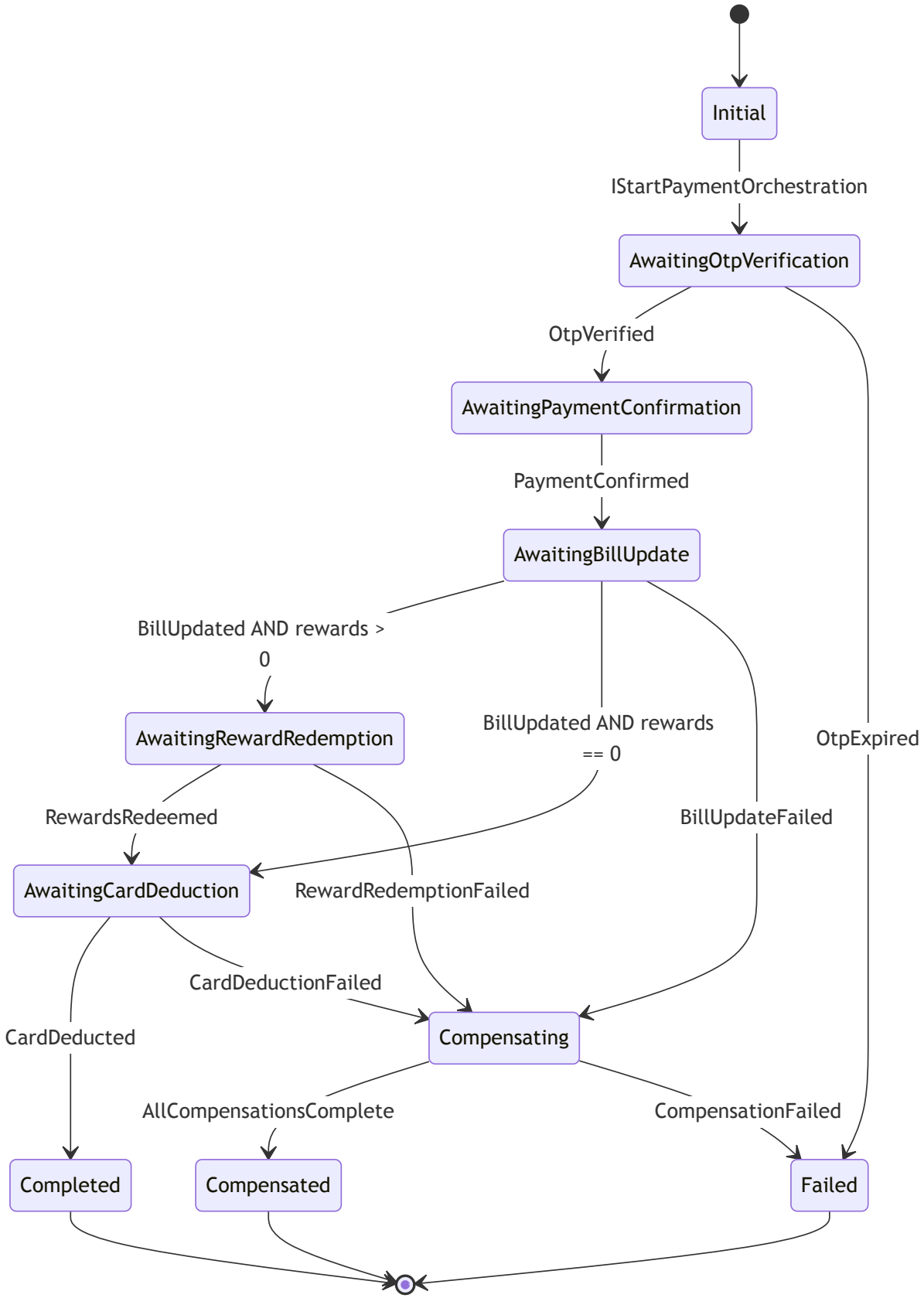
5.3 Saga Compensation (Payment Failure)



5.4 Wallet Top-Up via Razorpay

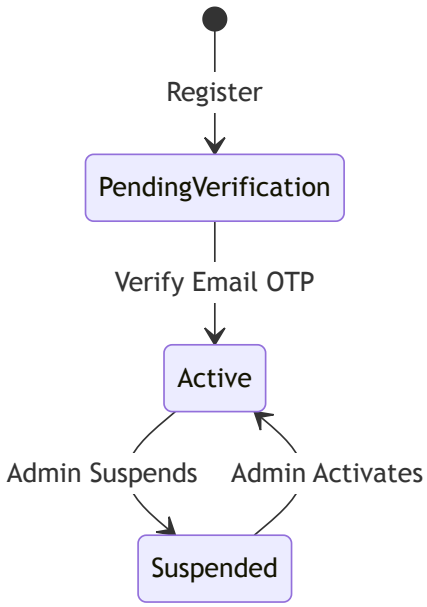


5.5 Payment Saga State Machine

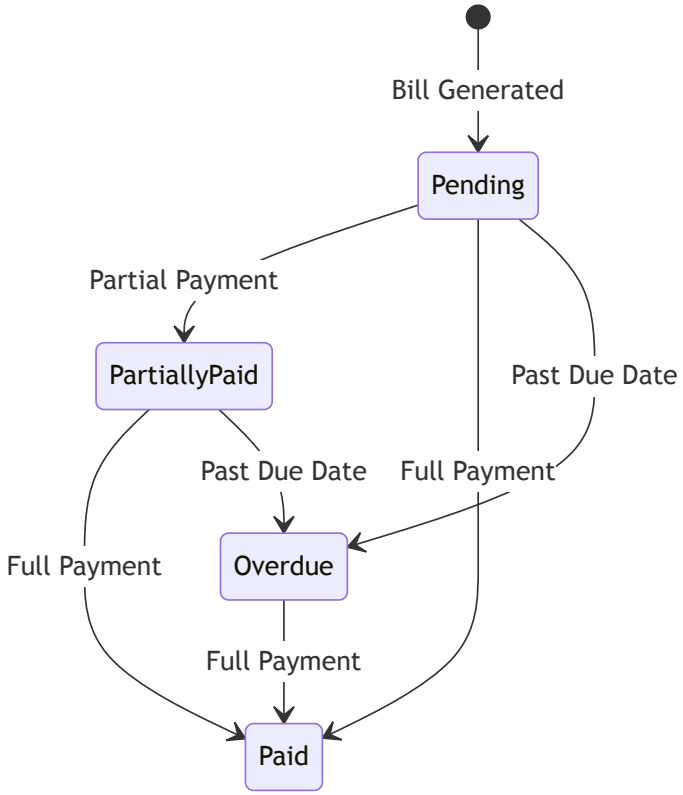


5.6 Entity State Machines

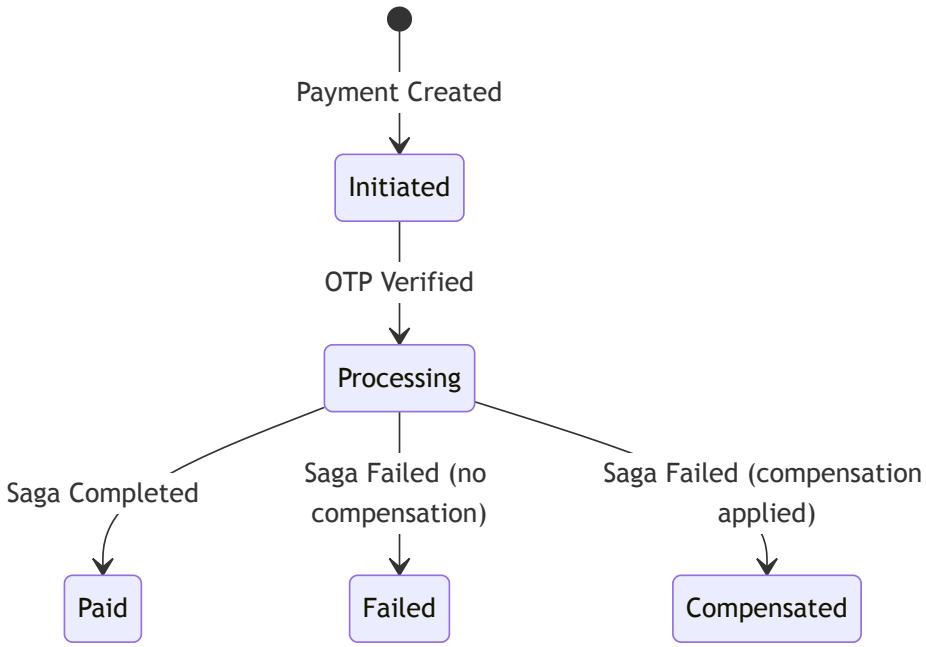
User Status:



Bill Status:



Payment Status:



6. Business Logic

6.1 User Rules

- Email must be unique across all users
- Password minimum: 8 characters, 1 uppercase, 1 lowercase, 1 digit
- OTP expires after 10 minutes (email verification/password reset), 5 minutes for payment OTP
- Suspended users cannot login or perform any action
- Google SSO users have null PasswordHash

6.2 Card Rules

- Card number validated via Luhn algorithm before encryption
- Card number encrypted with AES-256-CBC before storage
- Only last 4 digits stored in plain text (Last4 field)
- Masked number generated as `**** **** **** XXXX`
- Soft delete via `IsDeleted` flag with global query filter
- One card can be marked as default; setting a new default unsets the previous

6.3 Billing Rules

- Minimum due calculated as percentage of total bill amount
- Bills transition to Overdue if not paid by due date
- Partial payments allowed (status → PartiallyPaid)
- Statements aggregate all transactions within billing period

6.4 Rewards Rules

- Reward rate determined by tier (network + issuer specific)
- IssuerId on tier is nullable — null means applies to all issuers of that network
- Rewards earned on payment completion
- Rewards redeemable during payment (reduces payment amount)
- Reward redemption is reversible via saga compensation
- Each user has exactly one reward account

6.5 Payment Rules

- Payments require OTP verification (2FA)
- Payment via Wallet or Card (PaymentType enum)
- Wallet balance must be sufficient for wallet payments
- Expired payments cleaned up by background job
- All payments go through saga orchestration
- Payment amount must be >= minimum due

6.6 Wallet Rules

- Each user has exactly one wallet (auto-created on first use)
- Wallet balance cannot go negative
- Wallet transactions are immutable (append-only)
- Razorpay top-ups verified via HMAC-SHA256 signature
- Duplicate webhooks handled idempotently (check RazorpayPaymentId uniqueness)

6.7 Saga Rules

- Saga is idempotent (same CorrelationId = same saga instance)
- Compensation triggered on any step failure
- Compensation executes in reverse order of completed steps
- Saga state persisted in `PaymentOrchestrationSagas` table
- 30-second timeout per step triggers compensation

6.8 Core Algorithms

OTP Generation: Generate random 6-digit number (100000-999999), set expiry (10 min for email, 5 min for payment), store in record, send via notification.

Card Number Encryption: Encrypt plain number using AES-256-CBC → store in EncryptedCardNumber. Extract last 4 digits → Last4. Generate masked format → MaskedNumber. Never return plain number.

Reward Calculation: Find matching RewardTier by CardNetwork + IssuerId (fallback to network-only tier if IssuerId = null). Points = spendAmount × RewardRate. Add to RewardAccount.PointsBalance. Create RewardTransaction (Earned).

Razorpay Webhook Verification: Construct expected signature: HMAC-SHA256(orderId + "|" + paymentId, secret). Compare with received razorpay_signature. If match: update wallet, create transaction, mark top-up complete. If mismatch: reject 400. Check RazorpayPaymentId uniqueness for idempotency.

7. Error Handling

7.1 Error Response Format

```
{
  "success": false,
  "data": null,
  "message": "Validation failed",
  "errors": ["Email is required", "Password must be at least 8 characters"]
}
```

7.2 HTTP Status Codes

Status Code	Usage
200	Successful GET, PUT, PATCH
201	Successful POST (resource created)
400	Validation errors, invalid input
401	Missing or invalid JWT token
403	Insufficient permissions (role mismatch)
404	Resource not found
409	Conflict (duplicate email, duplicate card)
429	Rate limit exceeded
500	Internal server error

7.3 Validation Errors

Rule	Error Message
Email format	"Email must be a valid email address"
Email max length	"Email must not exceed 256 characters"
Password length	"Password must be at least 8 characters"
Password uppercase	"Password must contain at least one uppercase letter"
Password lowercase	"Password must contain at least one lowercase letter"
Password digit	"Password must contain at least one digit"
Full name min	"Full name must be at least 2 characters"

Rule	Error Message
Card number Luhn	"Card number is invalid"
Expiry date	"Card has expired"
Payment amount	"Payment amount must be greater than zero"
Wallet balance	"Insufficient wallet balance"

7.4 Retry Logic

Context	Strategy
MassTransit consumers	Exponential backoff: 1s → 5s → 15s (3 attempts)
External API calls (Razorpay, Google)	Not configured in current scope; should add Polly in production
Database operations	SQL Server handles transient faults natively

7.5 Failure Scenarios

Scenario	Handling
OTP expired	Return 400: "OTP has expired, please request a new one"
OTP mismatch	Return 400: "Invalid OTP, please try again"
Bill already paid	Return 400: "Bill is already paid"
Card not found	Return 404
Saga step timeout	Trigger compensation after 30 seconds
Razorpay webhook signature mismatch	Return 400, log security event
Gmail SMTP failure	Log error in NotificationLog, retry via MassTransit
Service unavailable	Gateway returns 502; other services continue operating

8. Security Implementation

8.1 Password Hashing

- **Algorithm:** BCrypt via `BCrypt.Net-Next` library
- **Work factor:** Default (10 rounds)
- **SSO users:** PasswordHash is null for Google OAuth users

8.2 JWT Token Management

Token Payload:

```
sub (UserId), email, name, role, iss (CredVault.Identity), aud (CredVault), exp, iat
```

- **Issuer:** Identity Service (shared secret across all services)
- **Expiry:** 60 minutes
- **Clock skew:** 30 seconds tolerance
- **Client storage:** `sessionStorage` (cleared on tab close)
- **Auto-attachment:** Angular HTTP interceptor adds Bearer token to all requests
- **401 handling:** Interceptor redirects to login on 401 response

8.3 Card Number Encryption

- **Algorithm:** AES-256-CBC
- **Key management:** Service-level configuration (should use Azure Key Vault / AWS KMS in production)
- **Fields stored:** EncryptedCardNumber (encrypted), Last4 (plain), MaskedNumber (masked)
- **Never exposed:** Plain card number is never returned in any API response

8.4 Role-Based Access Control (RBAC)

Attribute	Access
`[AllowAnonymous]`	Registration, login, OTP verification, password reset, Razorpay webhook
`[Authorize(Roles = "User")]`	Own cards, bills, payments, wallet, profile
`[Authorize(Roles = "Admin")]`	User management, card management, bill generation, reward tiers, logs

8.5 Razorpay Webhook Security

- **Signature verification:** HMAC-SHA256 of `orderId + "|" + paymentId` using webhook secret
- **Idempotency:** RazorpayPaymentId stored with UNIQUE constraint; duplicate webhooks return 200 without reprocessing
- **Endpoint:** Only webhook endpoint is `[AllowAnonymous]`; all other payment endpoints require auth

8.6 CORS Configuration

- Angular frontend origin allowed in all services
- Configured via `appsettings.json` per environment

9. Performance Optimizations

9.1 Query Optimization

- **Indexed foreign keys:** `UserId` on CreditCards, `CardId` on CardTransactions, `BillId` on Statements
- **Unique constraints:** `Email` on IdentityUsers, `UserId` on UserWallets and RewardAccounts
- **Soft-delete filter:** Global query filter (`IsDeleted == false`) avoids scanning deleted records
- **Pagination:** All list endpoints support Page/PageSize to limit result sets

9.2 CQRS Benefits

- Commands and queries separated, allowing independent optimization
- Read queries don't acquire write locks
- Future-ready for read-model caching (Redis) without affecting write path

9.3 Async Communication

- Non-critical operations (emails, audit logs) handled asynchronously via RabbitMQ
- Payment saga runs async — user receives immediate "Processing" response
- Notification service consumption doesn't block request-response flow

9.4 Lazy Loading Considerations

- EF Core lazy loading disabled by default
- Related data loaded explicitly via `.Include()` to avoid N+1 queries
- Statement transactions loaded separately from statement header

9.5 Background Jobs

- `PaymentExpirationJob` : Cleans up expired unpaid payments (runs periodically)
- `CheckOverdueBillsCommand` : Marks bills past due date as Overdue (admin-triggered)

10. Logging & Monitoring

10.1 Logging Strategy

- **Framework:** Serilog with structured logging
- **Output:** Rolling daily files (per service)
- **Log levels:** Information (default), Warning, Error, Fatal
- **Format:** JSON-structured for machine parsing

10.2 Correlation IDs

- Every incoming request generates a `TraceId`

- TraceId passed through HTTP headers between services, RabbitMQ message headers, database audit logs, and notification logs
- Enables end-to-end request tracing across all services

10.3 Audit Trail

- **Entity:** `AuditLog` in Notification Service
- **Captured:** EntityName, EntityId, Action, UserId, Changes (JSON), TraceId
- **Triggered:** All admin actions, user profile changes, card updates, payment state changes
- **Queryable:** Admin endpoint `/api/v1/notifications/audit`

10.4 Notification Logging

- **Entity:** `NotificationLog` in Notification Service
- **Captured:** UserId, Recipient, Subject, Body, Type, IsSuccess, ErrorMessage, TraceId
- **Purpose:** Track email delivery success/failure for debugging
- **Queryable:** Admin endpoint `/api/v1/notifications/logs`

10.5 Metrics (Recommended for Production)

- Request latency per endpoint, error rates per service, saga completion vs compensation rates, RabbitMQ queue depths
- **Tools:** Prometheus + Grafana (infrastructure), ELK Stack (logs)

11. Configuration Management

11.1 Environment Variables

Variable	Service	Purpose
`ConnectionStrings__DefaultConnection`	All services	Database connection string
`Jwt__Key`	All services	JWT signing/verification key
`Jwt__Issuer`	Identity Service	JWT issuer claim
`Jwt__Audience`	All services	JWT audience claim
`Jwt__ExpiryMinutes`	Identity Service	Token expiry duration
`Google__ClientId` / `Google__ClientSecret`	Identity Service	Google OAuth credentials
`RabbitMQ__Host` / `RabbitMQ__Username` / `RabbitMQ__Password`	All services	RabbitMQ connection

Variable	Service	Purpose
`Smtp__Host` / `Smtp__Port` / `Smtp__Username` / `Smtp__Password`	Notification Service	Gmail SMTP credentials
`Razorpay__KeyId` / `Razorpay__KeySecret` / `Razorpay__WebhookSecret`	Payment Service	Razorpay API credentials
`CardEncryption__Key`	Card Service	AES encryption key

11.2 appsettings.json Structure

Each service maintains its own `appsettings.json` with:

- ConnectionStrings (database)
- JWT configuration (shared secret, issuer, audience, expiry)
- RabbitMQ connection details
- Serilog logging configuration (console + rolling file, JSON format)
- Service-specific settings (SMTP, Razorpay, Google OAuth, CardEncryption)

11.3 Secrets Handling

- **Development:** `appsettings.Development.json` (gitignored)
- **Docker Compose:** Environment variables in `docker-compose.yml`
- **Production (recommended):** Azure Key Vault / AWS Secrets Manager
- **NEVER commit:** JWT keys, SMTP passwords, Razorpay secrets, encryption keys

12. Test Strategy

12.1 Unit Tests

Scope	What to Test	Tools
Command Handlers	Business logic, state changes, event publishing	xUnit, Moq, FluentAssertions
Query Handlers	Data retrieval, filtering, pagination	xUnit, Moq, FluentAssertions
Validators	Input validation rules, edge cases	FluentValidation.TestHelper
Domain Logic	Entity behavior, business rules, state transitions	xUnit
Saga State Machine	State transitions, event handling, compensation	MassTransit Test Harness

Key test scenarios: valid registration creates user and publishes events, login rejects unverified/suspended accounts, card addition encrypts number and publishes event, OTP validation rejects expired/incorrect codes.

12.2 Integration Tests

Scope	What to Test	Tools
API Endpoints	Full request/response cycle with test database	WebApplicationFactory, Testcontainers
Database Operations	EF Core queries, migrations, relationships	Testcontainers (SQL Server)
MassTransit Consumers	Event consumption, side effects	MassTransit InMemory Test Harness

Key test scenarios: POST /auth/register returns 201 with userId, POST /cards returns 201 with masked card number, GET /bills returns paginated results, saga happy path completes all steps.

12.3 Contract Tests

Scope	What to Test	Tools
Event Schemas	Event interfaces match between publisher/consumer	Shared.Contracts library
API Responses	Response envelope consistent across services	Shared.BaseApiController
DTO Validation	Request DTOs validate correctly	FluentValidation tests

12.4 Saga Tests

Scope	What to Test	Tools
Happy Path	Full saga execution: bill update → reward redemption → card deduction	MassTransit Test Harness
Compensation	Step failure triggers reverse compensation	MassTransit Test Harness
Timeout	Step timeout triggers compensation	MassTransit Test Harness
Idempotency	Duplicate events don't break saga	MassTransit Test Harness

12.5 Edge Cases to Cover

Scenario	Expected Behavior
Duplicate email registration	Return 409 Conflict
Login with unverified account	Return 400 "verify email first"
Login with suspended account	Return 403 "account suspended"
Add card with invalid number (Luhn fail)	Return 400 validation error
Add card with past expiry	Return 400 validation error

Scenario	Expected Behavior
Payment with insufficient wallet balance	Return 400 "insufficient balance"
Payment on already-paid bill	Return 400 "bill already paid"
Expired OTP submission	Return 400 "OTP expired"
Razorpay webhook with wrong signature	Return 400, log security event
Duplicate Razorpay webhook	Return 200 (idempotent), no double-credit
Saga step failure mid-execution	Compensation reverses completed steps

End of Low-Level Design Document